

	Horizon Campus Faculty of Information Technology BSc (Hons.) in Information Technology/BIT (Hons.) in Networking and Mobile Computing
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Please fully complete all sections as requested.

Degree Program: BSc (Hons.) in Information Technology		
Research Details		
Student Details / Group Name:		
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Coordinating supervisor		
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Proposed Research Project Details:
1. Working Title
Design and Evaluation of an AI-Based Virtual Driving License System for Driver Identification and Predictive Traffic Law Enforcement in Sri Lanka
2. Problem Statement
Sri Lanka's traffic law enforcement and licensing processes currently depend on physical driving license cards and manual checks by officers. These traditional methods are vulnerable to forgery, loss, and inefficiencies in verifying identity or tracking violations. While countries like China and India have begun integrating digital and AI technologies in transport enforcement, Sri Lanka lacks a virtual license system with predictive and real-time capabilities. Existing systems do not support behavior-based risk analysis, real-time violation detection, or automated driver profiling. There is no intelligent, AI-powered solution in Sri Lanka that offers secure virtual licensing, predictive behavior analysis, or enhanced situational awareness for traffic officers. Addressing this gap can significantly improve road safety, reduce law enforcement delays, and support Sri Lanka's digital transformation goals. iPermit, with its features such as facial recognition, danger zone alerts, white-line crossing detection, and accident visualization, aims to fill this critical need.
3. Research Aim
To design, develop, and evaluate an AI-powered virtual driving license system that enables real-time driver identification through facial recognition, facilitates digital access to driving history and traffic violations, and enhances road safety in Sri Lanka through intelligent features such as accident detection, white-line crossing recognition, behavioral tracking, a point-based violation system, and performance-based driver rewards.

4. Research Objectives

Objective 1: To implement and evaluate a facial recognition system for accurate and real-time authentication of drivers, using AI-based algorithms under realistic traffic conditions.

Objective 2: To develop and integrate intelligent traffic monitoring features—including white-line violation detection, accident location visualization, and a point-based penalty system—into the virtual license application.

Objective 3: To assess the predictive capabilities of AI in tracking driver behavior patterns and determining point deductions or license suspensions based on past violations.

Objective 4: To design and conduct usability studies with traffic officers and licensed drivers in order to evaluate the practicality, efficiency, and user acceptance of the proposed virtual license system.

5. Research Questions

Question 1: How accurately and reliably can facial recognition technology be applied to authenticate driver identities and retrieve virtual license data in real-time traffic law enforcement scenarios in Sri Lanka?

Question 2: What is the effectiveness of AI-powered modules—such as white-line violation detection, accident localization, and behaviour-based point deduction—in enhancing proactive traffic monitoring and improving road safety outcomes?

Question 3: How do end-users, including licensed drivers and traffic enforcement personnel, perceive the usability, efficiency, and overall impact of transitioning from traditional physical licenses to an AI-enabled virtual driving license system?

6. Background Literature Review

Traffic law enforcement remains a critical issue in developing countries like Sri Lanka, where traditional driving license systems depend heavily on physical cards and manual verification. These outdated processes are often inefficient, prone to forgery, and lack real-time capabilities. As urban populations grow and traffic volume increases, law enforcement

authorities face new challenges in ensuring road safety and maintaining compliance with traffic regulations. Consequently, there is an increasing demand for digital, intelligent systems that can support traffic policing and driver monitoring with higher accuracy and lower operational costs.

Artificial Intelligence (AI) and facial recognition technologies have shown significant potential in automating identity verification and enhancing the security of public systems. According to Zhao et al. (2019), facial recognition systems achieve over 95% accuracy in real-time conditions when optimized with convolutional neural networks (CNNs), making them highly suitable for applications such as license verification. Facial recognition has been successfully integrated into smart policing systems in countries like China and India, offering a secure, contactless way to identify individuals (Wang & Wu, 2020).

Moreover, traffic safety researchers have begun using computer vision and machine learning to detect specific violations, such as red-light running and white-line crossing. Mandal et al. (2021) developed an AI-powered camera system to detect traffic violations with 88% accuracy, demonstrating how similar methods could be adopted in Sri Lanka to improve enforcement.

Another promising trend is the use of AI to predict future violations based on past driver behavior. Predictive analytics allows for preemptive action, such as early warnings or temporary license suspensions. Cheng et al. (2020) found that behavior-based prediction models could anticipate reckless driving incidents with high precision when trained on large-scale driving datasets.

The concept of a digital, point-based license system has also gained traction. European countries such as Germany and France use systems where points are deducted for traffic violations, and licenses are suspended once a threshold is reached. Integrating such systems into mobile applications, as suggested by Silva (2022), promotes accountability and rewards responsible driving. Your proposed system, "iPermit," adopts a similar approach with point tracking, suspension triggers, and reward features for top drivers, making it highly relevant and timely.

Additionally, real-time accident location detection and visualization have become essential tools in modern traffic systems. Using GPS and AI-based mapping, authorities can better manage traffic flow and emergency responses. Google Maps and Waze have proven the effectiveness of user-driven and automated incident reporting features.

Despite these advancements globally, Sri Lanka lacks an integrated virtual license platform that combines real-time AI facial recognition, behavior tracking, point-based penalties, and predictive safety monitoring. Government digitalization efforts are still in early stages, and existing enforcement mechanisms remain reactive rather than proactive.

Therefore, this research and product development are highly justified. It aims to bridge the current technological gap by building a comprehensive, AI-powered virtual license system tailored to Sri Lanka's enforcement landscape.

7. Data Requirement

A: Primary Data

Primary data will be collected through user testing, surveys, and structured interviews.

- **Facial Recognition Testing:** Facial images from volunteer participants will be captured to train and evaluate the performance of the AI-based face recognition module.
- **User Surveys & Interviews:** Questionnaires will be distributed to licensed drivers and traffic police officers to gather feedback on the usability, trust, and practicality of the virtual license system.
- **Behavior Analysis Simulations:** Driving behavior (e.g., violation patterns, point deductions) will be simulated in controlled environments to test the effectiveness of the point-based system and prediction features.

B: Secondary Data

Secondary data will be obtained from publicly available datasets, technical reports, and government sources:

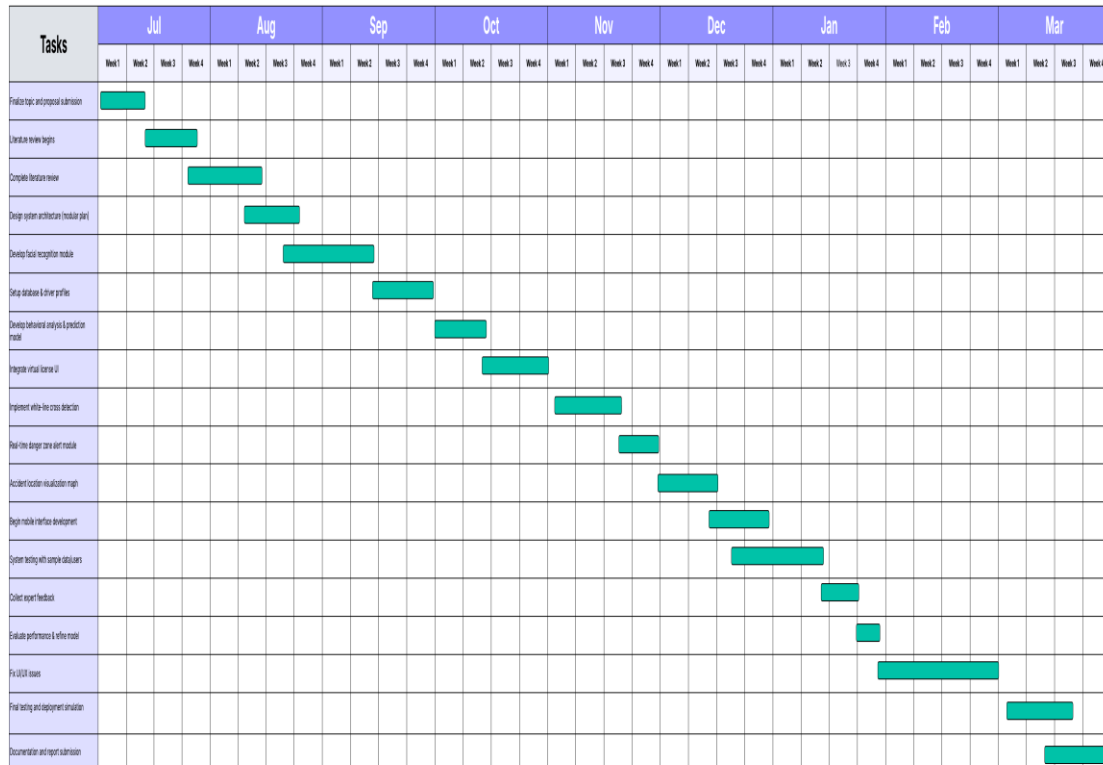
- **Traffic Violation Records & Accident Reports** from the Sri Lankan Ministry of Transport.
- **Literature Reviews and Case Studies** on AI in traffic enforcement, facial recognition systems, and smart transportation solutions from journals, conference papers, and research repositories.

8. Hardware/ Software Requirements Software Requirements: • Flutter • React.js / MongoDB • Python • Firebase / Supabase (Optional) • Pandas • Scikit learn • FastAPI • Postman / GitHub / Figma • FaceAPI.js / TensorFlow.js • OpenCV + YOLOv5 for dashcam AI Hardware Requirements: • Smartphone with dashcam (for simulation)
9. Ethics Yes, this research may involve ethical considerations related to the collection and use of personal biometric data, such as facial images and driving behavior patterns. Since facial recognition technology is used to verify driver identity, informed consent will be obtained from all participants involved in data collection and system testing. Measures will be taken to anonymize and securely store all collected data to prevent misuse or privacy breaches. No sensitive information will be shared publicly, and all testing will be conducted in a controlled environment with voluntary participants. The system will be designed with privacy and data protection principles in mind, ensuring compliance with local data protection regulations. Ethical approval will be sought prior to any human data collection, and participants will be fully informed of their rights, including the right to withdraw from the study at any time.

10. References

1. W. Zhao, R. Chellappa, P. J. Phillips, and A. Rosenfeld, "Face recognition: A literature survey," *ACM Computing Surveys*, vol. 35, no. 4, pp. 399–458, 2019.
2. L. Cheng, Z. Wang, and H. Liu, "Predictive analytics in traffic safety," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 4, pp. 1418–1430, Apr. 2020.
3. V. Mandal, S. Sharma, and K. Patel, "AI-Driven Traffic Monitoring System for Smart Cities," *Proc. IEEE Conf. Smart Technologies*, pp. 115–120, 2021.
4. N. D. Silva, "Digital Transformation in Public Services in Sri Lanka," *ICTA Report*, 2022.
5. J. Wang and H. Wu, "Smart Policing with Facial Recognition: A Case Study in Shenzhen," *International Journal of Surveillance Tech.*, vol. 8, no. 1, pp. 33–45, 2020.
6. "What is facial recognition technology?," Algorithmic Justice League. [Online]. Available: <https://www.ajl.org>.
7. "The Impact of Biometric Surveillance on Reducing Violent Crime," *PMC*, Jun. 2021.
8. "A predictive state? Sri Lankan context of predictive policing," *Daily FT*, Colombo, 2020.
9. Ministry of Transport, Sri Lanka, "Annual Road Accident Report 2023," *Gov.lk*, 2023.
10. World Bank, "Smart Transport Systems in Developing Countries," *Transport Review Report*, 2021.

11. Gantt Chart



https://lucid.app/lucidchart/e7ef5660-048d-40b8-a6d5-c8354c6e1572/edit?view_items=HOZuxlRBGZfT&invitationId=inv_87b789bc-42e4-426e-97c1-cd651a25b4cb

Students Agreement:		
By signing below on <u>06/07/2025</u>, We, as B.R.G.S Sandaruwan, J.P.I.S Jayasinghe and B.R Vindyani agree to the commonly known and accepted research ethics and guidelines. We commit to conducting the research project as mentioned above and outlined in the agreement. We understand that any changes to the group composition, identified research problem, or selected supervisor may result in penalties.		
Name with Initials	Registration No	Signature
B.R.G.S Sandaruwan	ITBIN-2211-0278	
J.P.I.S Jayasinghe	ITBIN-2211-0324	
B.R Vindyani	ITBIN-2211-0122	
Supervisor(s) Agreement:		
By signing below, I/We agree to supervise the aforementioned group of students throughout their research project, providing guidance and support for successful completion. Furthermore, by signing this document, I/we give my/our consent to the following: 1. The identified research problem is justifiable and feasible for investigation. 2. The research aim is achievable within a one-year timeframe. 3. The specified research objectives are measurable and achievable within the research period.		
Supervisor's Name with Initials	Signature	Date
Departmental Use Only:		

Date Proposal Form Submitted: _____

All required sections of the form have been accurately completed, and the necessary signatures have been placed. The initial research proposal has been partially approved, thereby allowing the proposed research project to proceed in accordance with the specified guidelines outlined in the IT41028.

Research Coordinator's Signature:_____ **Date:** _____

Department Head's Signature:_____ **Date:** _____

